

packages

- esc+m ===== code mode to markdown
- esc+y ===== markdown to code mode
- shift+enter

```
In [ ]: Machine learning ===== sickit learn
        Deep learning ===== Tensorflow
        NLP ===== NLTK
```

```
In [ ]: addition of two numbers logic
```

```
a=10
b=20
c=a+b
c
```

```
In [ ]: you went courier shop to deliver that
        anand kumar alos packages that code ===== python organization
                                                ===== anaconda organization

        they made one agreement
```

```
In [ ]: Q) generate a random number between 10 and 20
```

```
# you no need to develope a code

alex ===== package ===== random

installed anaconda ===== random package will be there in your local
```

```
In [ ]: # step-1:
        import <package name>
```

```
In [1]: import random

        # if it is there means it will run
        # otherwise you will get an error
        # module not found error
```

```
In [2]: random
```

```
Out[2]: <module 'random' from 'C:\\Users\\omkar\\anaconda3\\Lib\\random.py'>
```

```
In [ ]: import cv2 ===== error

        cv : computer vision package
```

```
In [ ]: anaand created a addition ===== package ==== anaconda
        subtraction
        multiplication
        division

package: math operations
```

```
In [ ]: import math operations
```

```
In [ ]: # step-2:

in the random package how many sub modules are there
In the random pcakage how many methods are avialable

dir(<package_name>)
```

In [3]: `dir(random)`

```
Out[3]: ['BPF',
         'LOG4',
         'NV_MAGICCONST',
         'RECIP_BPF',
         'Random',
         'SG_MAGICCONST',
         'SystemRandom',
         'TWOPI',
         '_ONE',
         '_Sequence',
         '_Set',
         '__all__',
         '__builtins__',
         '__cached__',
         '__doc__',
         '__file__',
         '__loader__',
         '__name__',
         '__package__',
         '__spec__',
         '_accumulate',
         '_acos',
         '_bisect',
         '_ceil',
         '_cos',
         '_e',
         '_exp',
         '_floor',
         '_index',
         '_inst',
         '_isfinite',
         '_log',
         '_os',
         '_pi',
         '_random',
         '_repeat',
         '_sha512',
         '_sin',
         '_sqrt',
         '_test',
         '_test_generator',
         '_urandom',
         '_warn',
         'betavariate',
         'choice',
         'choices',
         'expovariate',
         'gammavariate',
         'gauss',
         'getrandbits',
         'getstate',
         'lognormvariate',
         'normalvariate',
         'paretovariate',
         'randbytes',
         'randint',
         'random',
         'randrange',
         'sample',
         'seed',
         'setstate',
```

```
'shuffle',  
'triangular',  
'uniform',  
'vonmisesvariate',  
'weibullvariate']
```

```
In [4]: # step-1: import <package_name>
# step-2: dir(<pacakage_name>)
import random
dir(random)
```

```
Out[4]: ['BPF',
         'LOG4',
         'NV_MAGICCONST',
         'RECIP_BPF',
         'Random',
         'SG_MAGICCONST',
         'SystemRandom',
         'TWOPI',
         '_ONE',
         '_Sequence',
         '_Set',
         '__all__',
         '__builtins__',
         '__cached__',
         '__doc__',
         '__file__',
         '__loader__',
         '__name__',
         '__package__',
         '__spec__',
         '_accumulate',
         '_acos',
         '_bisect',
         '_ceil',
         '_cos',
         '_e',
         '_exp',
         '_floor',
         '_index',
         '_inst',
         '_isfinite',
         '_log',
         '_os',
         '_pi',
         '_random',
         '_repeat',
         '_sha512',
         '_sin',
         '_sqrt',
         '_test',
         '_test_generator',
         '_urandom',
         '_warn',
         'betavariate',
         'choice',
         'choices',
         'expovariate',
         'gammavariate',
         'gauss',
         'getrandbits',
         'getstate',
         'lognormvariate',
         'normalvariate',
         'paretovariate',
         'randbytes',
         'randint',
         'random',
         'randrange',
         'sample',
         'seed',
         'setstate',
```

```
'shuffle',
'triangular',
'uniform',
'vonmisesvariate',
'weibullvariate']
```

In [5]: `a=10 # it has ran`

In [6]: `name`

```
-----
-
NameError                                Traceback (most recent call las
t)
Cell In[6], line 1
----> 1 name

NameError: name 'name' is not defined
```

In []: `randint`

you have a friend : `randint===== random`

In []: `random has many kids in that one kid is randint`

In []: `<package_name>.<kidname>`

In [7]: `help(random.randint)`

Help on method randint in module random:

randint(a, b) method of random.Random instance
Return random integer in range [a, b], including both end points.

In [8]: `# generate a random number between 10 and 20
step-1: import <package_name>
step-2: dir(<package_name>)
step-3: help(<package_name>.<method_name>)`

```
import random
dir(random)
help(random.randint)
```

Help on method randint in module random:

randint(a, b) method of random.Random instance
Return random integer in range [a, b], including both end points.

In [15]: `random.randint(10,20)`

Out[15]: 19


```
In [17]: # generate a random number between 10 and 20
# step-1: import <package_name>
# step-2: dir(<package_name>)
# step-3: help(<package_name>.<method_name>)
# step-4: <package_name>.<method_name>()

import random
dir(random)
help(random.randint)
random.randint(10,30)
```

Help on method randint in module random:

randint(a, b) method of random.Random instance
Return random integer in range [a, b], including both end points.

Out[17]: 28

```
In [19]: # Package name : math  
# do the first 3 steps  
# pi  
  
import math  
dir(math)
```

```
Out[19]: ['__doc__',
          '__loader__',
          '__name__',
          '__package__',
          '__spec__',
          'acos',
          'acosh',
          'asin',
          'asinh',
          'atan',
          'atan2',
          'atanh',
          'cbrt',
          'ceil',
          'comb',
          'copysign',
          'cos',
          'cosh',
          'degrees',
          'dist',
          'e',
          'erf',
          'erfc',
          'exp',
          'exp2',
          'expm1',
          'fabs',
          'factorial',
          'floor',
          'fmod',
          'frexp',
          'fsum',
          'gamma',
          'gcd',
          'hypot',
          'inf',
          'isclose',
          'isfinite',
          'isinf',
          'isnan',
          'isqrt',
          'lcm',
          'ldexp',
          'lgamma',
          'log',
          'log10',
          'log1p',
          'log2',
          'modf',
          'nan',
          'nextafter',
          'perm',
          'pi',
          'pow',
          'prod',
          'radians',
          'remainder',
          'sin',
          'sinh',
          'sqrt',
          'tan',
```

```
'tanh',  
'tau',  
'trunc',  
'ulp']
```

In [20]: `help(math.pi)`

Help on float object:

```
class float(object)  
| float(x=0, /)  
|  
| Convert a string or number to a floating point number, if possible.  
|  
| Methods defined here:  
|  
| __abs__(self, /)  
|     abs(self)  
|  
| __add__(self, value, /)  
|     Return self+value.  
|  
| __bool__(self, /)  
|     True if self else False  
|  
| __ceil__(self, /)  
|     Return the ceiling of an Integral
```

In [21]: `math.pi`

Out[21]: 3.141592653589793

In [22]: `import math
dir(math)
math.pi`

Out[22]: 3.141592653589793

In [23]: `help(math.sin)`

Help on built-in function sin in module math:

```
sin(x, /)  
    Return the sine of x (measured in radians).
```

In [25]: `math.sin(90)`

Out[25]: 0.8939966636005579

In [26]: `import math
math.sin(90)`

Out[26]: 0.8939966636005579

```
In [ ]: # sqrt
```

```
In [27]: help(math.sqrt)
```

Help on built-in function sqrt in module math:

```
sqrt(x, /)
    Return the square root of x.
```

```
In [30]: math.sqrt(64)
```

```
Out[30]: 8.0
```

```
In [ ]: import <package_name>
dir(<package_name>) # List of method
help(<package_name>.<method_name>)
<package_name>.<method_name>
```

```
In [ ]: package name: keyword
method name: kwlist

import <package_name>
dir(<package_name>) # kwlist
<package_name>.<method_name>
```

```
In [31]: import keyword
```

```
In [32]: dir(keyword)
```

```
Out[32]: ['__all__',
          '__builtins__',
          '__cached__',
          '__doc__',
          '__file__',
          '__loader__',
          '__name__',
          '__package__',
          '__spec__',
          'iskeyword',
          'issoftkeyword',
          'kwlist',
          'softkwlist']
```

```
In [35]: len(keyword.kwlist)
```

```
Out[35]: 35
```

```
In [34]: keyword
```

```
Out[34]: <module 'keyword' from 'C:\\\\Users\\omkar\\anaconda3\\Lib\\keyword.py'>
```

```
In [36]: import random
```

```
In [37]: random
```

```
Out[37]: <module 'random' from 'C:\\Users\\omkar\\anaconda3\\Lib\\random.py'>
```

In [38]: `dir(math)`

```
Out[38]: ['__doc__',
          '__loader__',
          '__name__',
          '__package__',
          '__spec__',
          'acos',
          'acosh',
          'asin',
          'asinh',
          'atan',
          'atan2',
          'atanh',
          'cbrt',
          'ceil',
          'comb',
          'copysign',
          'cos',
          'cosh',
          'degrees',
          'dist',
          'e',
          'erf',
          'erfc',
          'exp',
          'exp2',
          'expm1',
          'fabs',
          'factorial',
          'floor',
          'fmod',
          'frexp',
          'fsum',
          'gamma',
          'gcd',
          'hypot',
          'inf',
          'isclose',
          'isfinite',
          'isinf',
          'isnan',
          'isqrt',
          'lcm',
          'ldexp',
          'lgamma',
          'log',
          'log10',
          'log1p',
          'log2',
          'modf',
          'nan',
          'nextafter',
          'perm',
          'pi',
          'pow',
          'prod',
          'radians',
          'remainder',
          'sin',
          'sinh',
          'sqrt',
          'tan',
```



```
'tanh',  
'tau',  
'trunc',  
'ulp']
```

```
In [ ]: # how to install the packages does not there in your local
```

```
In [40]: import cv2
```

- we can install using jupyter notebook
- we can install using anaconda command prompt(cmd) (Recommended)

```
In [41]: !pip install opencv-python
```

Requirement already satisfied: opencv-python in c:\users\omkar\anaconda3\lib\site-packages (4.8.1.78)

Requirement already satisfied: numpy>=1.21.2 in c:\users\omkar\anaconda3\lib\site-packages (from opencv-python) (1.24.3)

- what is package
- what are sub methods
- how to understand the packages
- how to import the packages
 - random
 - math
 - keyword
- how to install the package

```
In [ ]:
```

```
In [ ]:
```